

EXPLORATION BARRIER

Original Paper Author

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Problem/Objective

The Exploration Barrier

Reinforcement learning (RL) methods perform well when **environments provide frequent reward signals**. However, many real-world and benchmark tasks contain sparse or delayed rewards, giving agents little feedback.

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Intrinsic Rewards

In sparse-reward environments, agents receive little to no feedback.

To encourage exploration, an intrinsic reward component is added to the extrinsic reward.

BY RANDOM NETWORK

Poster Presentation – Group #8

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Methodology

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Results

Montezuma's Revenge:

RND established a new state-of-the-
consistently found over half the rooms (22
24) and occasionally completed the first
achieving scores over 17,500.

WORK DISTILLATION

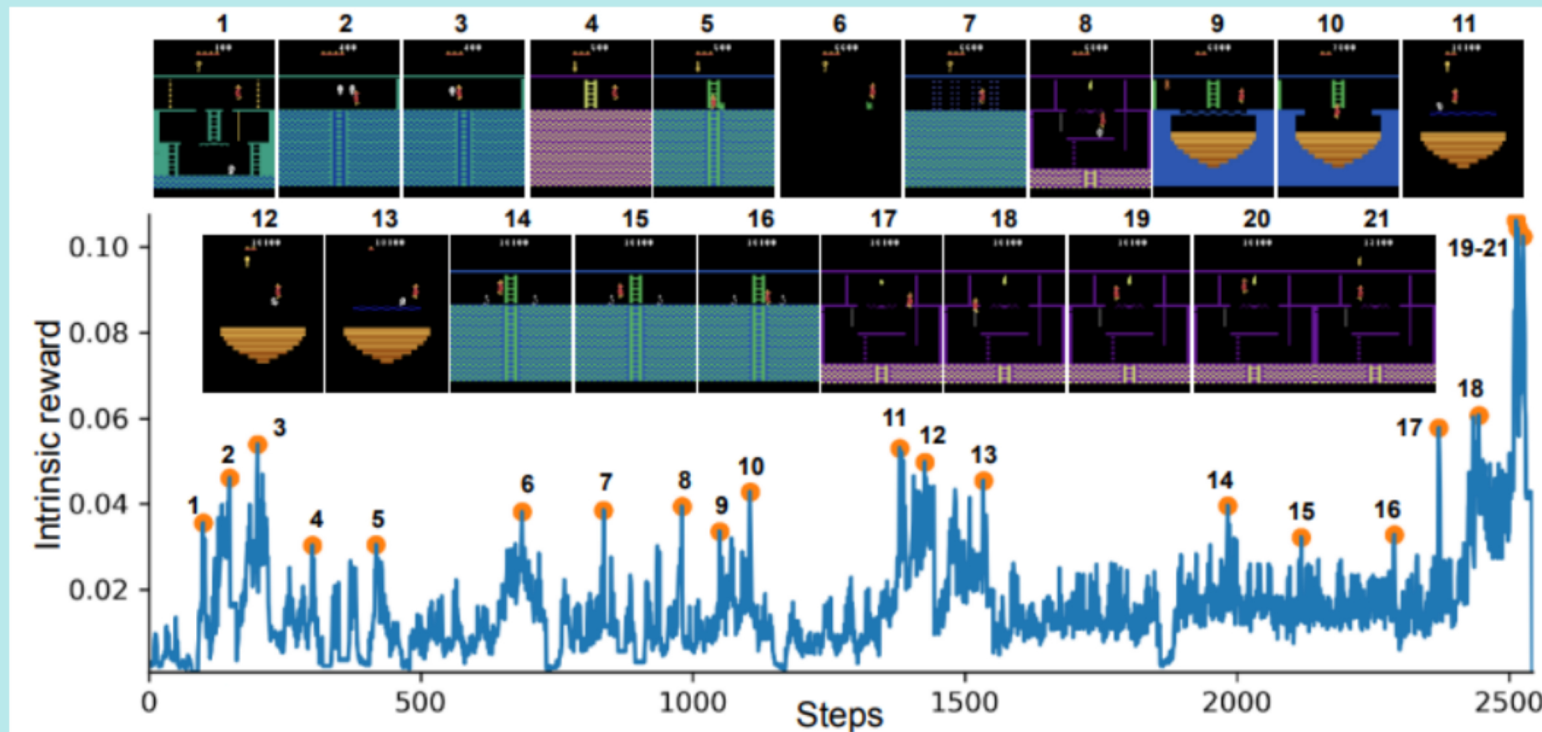
Group Members

Chavez Flores, Md Mosarraf Hossain, Mohammad

Limitations

- **Representation Sensitivity:** RND's novelty bonus depends heavily on the random feature space. Poor CNN features can make unrelated states appear similar, causing the agent to miss important distinctions.
- **Reward non-stationarity** : As the predictor improves, the intrinsic reward signal shrinks over time, which can

This makes **learning effective policies** through random exploration **extremely difficult**. As a result, agents often **fail to discover meaningful states or actions**, especially in complex environments like Montezuma's Revenge, where rewards require long sequences of correct actions. This exploration challenge limits the scalability of RL to large state spaces with sparse feedback.



Rewarding Novel States

component is added to the total reward. Total reward becomes the sum of extrinsic and intrinsic rewards.

$$\mathbf{r}_t = \mathbf{e}_t + \mathbf{i}_t$$

$\mathbf{i}_t$ is large for unfamiliar states and becomes smaller as states become familiar.

Random Network

Intrinsic reward is calculated as the difference between two network outputs.

- **Target Network** is fixed.
- **Predictor Network** is the target.

When encountering a new state, the predictor network produces a high intrinsic reward, which decreases as the state becomes familiar.

added to the reward signal.

It becomes the sum of extrinsic (e_t) and intrinsic (i_t) reward:

The reward is high for unfamiliar states and decreases as the state becomes more familiar.

Knowledge Distillation (RND)

The loss is computed as the MSE between the outputs of two neural networks:

Encoder network: Randomly initialized & frozen.

Decoder network: Tries to imitate the encoder.

When visiting a new state, the predictor makes a large error $\rightarrow$ Strong gradient. As a state is revisited, the error decreases.

Comparison to Baselines:

RND significantly outperformed standard baselines. "Dynamics-based" (forward-prediction) exploration performed well on games like Venture and Private Eye.

Pure Exploration:

Even when the game score was completely frozen, the RND agent successfully explored 21 rooms in Montezuma's Revenge just out of "curiosity".

Non-episodic intrinsic rewards improve exploration

When combining intrinsic + extrinsic, using non-episodic intrinsic (curiosity doesn't "die" once episode over) increases number of rooms explored (especially for RNN policies).

Scale helps (many parallel environments):

With more parallel environments (larger exploration batches), performance improves—policy learning is faster.

PPO and
Exploration

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Exploration:
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experience
earns to

destabilize training if not properly normalized.

- **No environment model** : RND has no understanding of why a state is novel, only that it is. It can get stuck exploring visually complex but strategically irrelevant states.

Why It Matters?

RND directly addresses one of the hardest open problems in deep RL: learning useful behavior when **rewards are extremely rare**. It achieves superhuman-level room exploration in Montezuma's Revenge – a benchmark that had stumped RL researchers for years – using a surprisingly simple mechanism. Beyond games, this has real implications for robotics, scientific discovery, and any domain where trial-and-error exploration is costly and feedback is sparse.

Conclusion

To address this challenge, the paper proposes **Random Network Distillation (RND)**, an intrinsic motivation method that encourages agents to **explore unfamiliar states**.

The approach introduces an exploration bonus based on prediction error between **two neural networks**: a fixed randomly initialized target network and a predictor network trained to imitate it.

When the agent encounters a new or rarely visited state, the predictor network produces a **large prediction error**, generating a **high intrinsic reward**. As the agent revisits the state, the prediction error decreases and the reward fades. This mechanism allows the agent to naturally **prioritize novel states**, improving exploration efficiency and enabling **learning in environments with sparse external rewards**.

Objective:

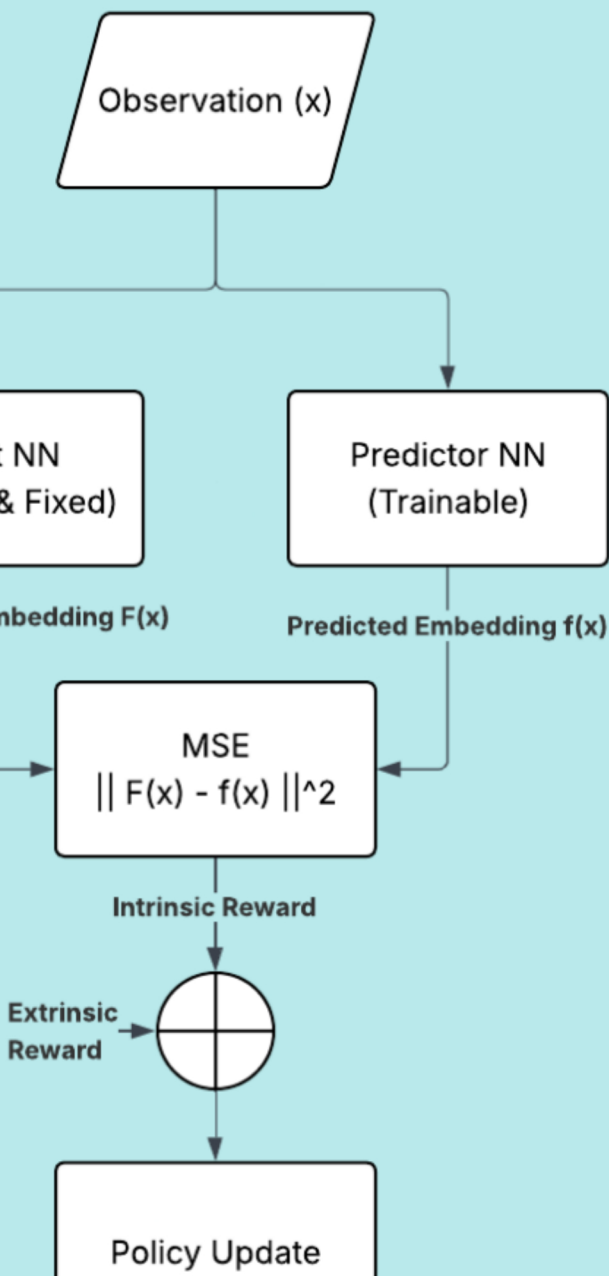
Develop a scalable exploration strategy that encourages agents to visit novel states using intrinsic rewards derived from prediction error.

Intrinsic reward.
predictor network
decreases, and the



As a state is revisited, the network learns them, the MSE and the reward decreases.

From Network Distillation Framework



exploit transient intrinsic rewards before they vanish. Paper shows very high returns and efficient exploration at scale; in some runs the agent explored all 24 rooms of level 1 and sometimes passed to the first level.

Qualitative behavior:

Agent sometimes seeks dangerous, rare states (jumping around hazards), such states are underrepresented in experience, so the agent is producing intrinsic reward.

Key Sources & Acknowledgements

- Burda, Y., Edwards, H., Storkey, A., & Klein, D. (2018). [arXiv](#)
- Schulman, J., et al. (2017). Proximal Policy Optimization
- Johnny Code. Reinforcement Learning Tutorial

Conclusion

RND offers an elegant, scalable solution to the hard exploration problem by repurposing **prediction error** as a **curiosity signal**. By keeping the target network fixed and random, it avoids the noisy-TV pitfall that undermines dynamics-based curiosity methods. The dual reward stream design – combining episodic extrinsic returns with non-episodic intrinsic returns – proves critical for crossing episode boundaries and achieving deep exploration. Its simplicity and compatibility with standard RL algorithms like PPO make it a strong baseline for any sparse-reward setting.

Acknowledgements

Timov, O. (2019). Exploration by Random Network Distillation. ICLR 2019.

Policy Optimization Algorithms. arXiv:1707.06347

Tutorials. YouTube. youtube.com/@johnnycode

